

METRO TICKET GENERATION SYSTEM

Overall Project Report

ServiceNow • Phases 1–5 • Requirements to Final Presentation

Consolidated project summary with implementation evidence and screenshots

1. Project Overview & End-to-End Architecture

The Metro Ticket Generation System is developed in ServiceNow as a phased solution that moves from requirement analysis and planning through backend configuration, passenger-facing UI/UX, testing and security, and final deployment/documentation.

Five-phase delivery structure

Phase	Primary focus	Outcome
1	Requirement Analysis & Planning	Approved blueprint, scope, MVP, rules and roadmap
2	Backend Development & Configurations	Data layer, tables, fields, relationships and master data
3	UI/UX Development & Customization	Booking catalog, dynamic fare logic and QR experience
4	Data Migration, Testing & Security	Validation, QA, ACL review, data integrity and field mapping
5	Deployment, Documentation & Final Presentation	Troubleshooting, documentation, demo and future roadmap

Architecture

User → Service Catalog → Variables → Client/UI Logic → Validation → Automation / Field Mapping → Backend Table → Ticket / QR Experience

Phase 3 connects the passenger-facing catalog to the backend foundation. Phase 4 adds validation, security and field mapping, while Phase 5 prepares the solution for deployment, handoff and presentation.

2. Phase 1 — Requirement Analysis & Planning

Phase 1 establishes what the system must achieve before development begins. Its focus is requirement discovery, clarification, prioritization and planning. The planned journey is Passenger → Journey Details → Validation → Fare Calculation → Review → Confirmation → Ticket Generation → Notification → Ticket History.

MVP scope

- Station selection and source/destination validation
- Journey date and passenger-count validation
- Fare calculation
- Booking creation and unique ticket generation
- Ticket status and ticket viewing
- Confirmation notification

Out of initial scope

Real-time train tracking and external transit APIs, online payment gateway and financial settlement, QR gate scanning / smart-card integration, dedicated mobile application and advanced SMS/push integrations, and AI-based route optimization or dynamic pricing were identified as future/out-of-scope items.

Planning risks and roadmap

Key risks include unclear fare policy, changing requirements, duplicate tickets, incorrect access, poor master data and integration dependencies. The planning material proposes business approval, scope baselining, unique identifiers, early ACL definition, master-data validation and explicit MVP boundaries.

Phase 2 Evidence — Metro Station Details Table Configuration

23. ServiceNow Implementation Evidence

The following screenshots are added to the Phase 2 document as direct implementation evidence for the Metro Station Details backend configuration. They complement the documented table-creation, field-creation and record-validation activities.

Metro Station Details — Table Configuration

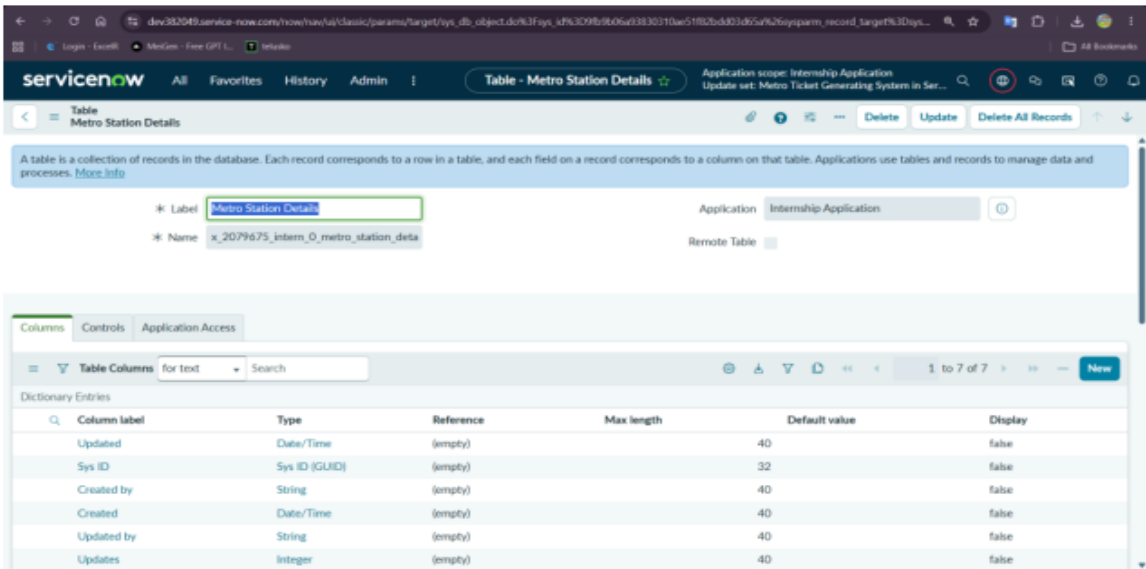


Figure 23. Metro Station Details table configuration showing the table definition and configured columns.

This evidence corresponds to the Phase 2 backend activity in which the Metro Station Details table is created as the master-data container for metro stations.

Phase 2 Evidence — Station Record Creation and Verification

24. Station Record Creation and Verification

New Station Record

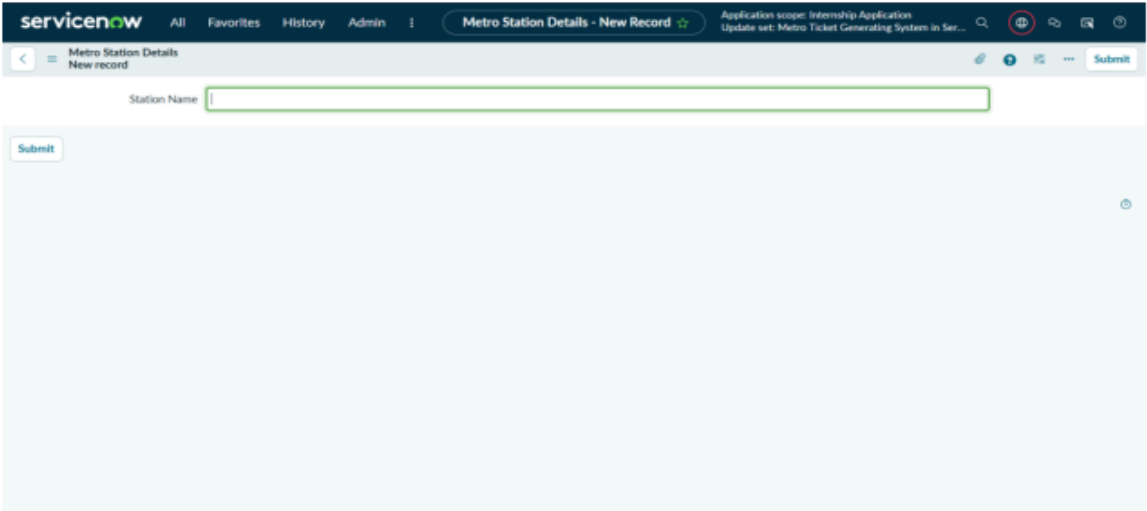
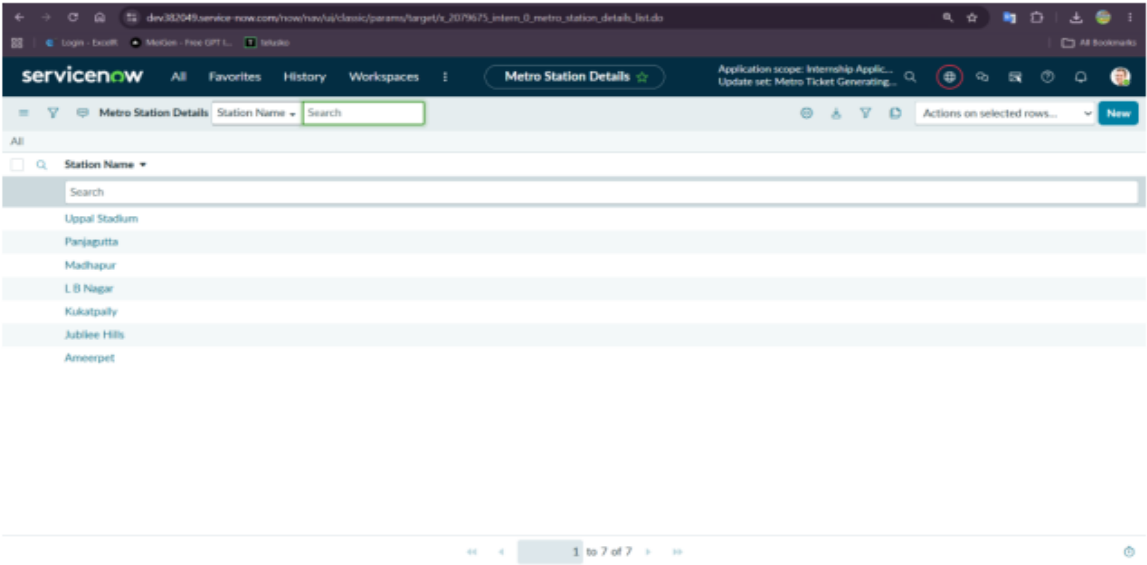


Figure 24. Metro Station Details — New Record form showing the Station Name field used for station master-data creation.

Station Records List



Station Name	Search
Uppal Stadium	
Panjagutta	
Madhapur	
L B Nagar	
Kukatpally	
Jubilee Hills	
Ameerpet	

Figure 25. Metro Station Details list showing the configured station records, including Uppal Stadium, Panjagutta, Madhapur, L B Nagar, Kukatpally, Jubilee Hills and Ameerpet.

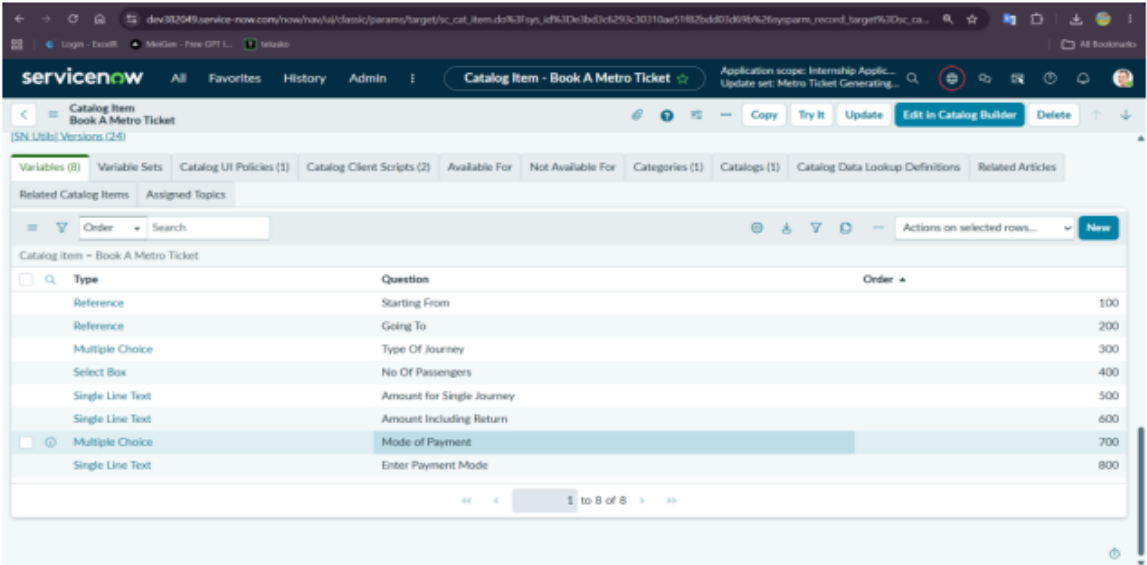
Together, these screens show the practical backend flow: the Station Name field is presented for record creation, and the saved station records can then be viewed in the Metro Station Details list. This supports the Phase 2 testing objective that valid station records can be created and retrieved successfully.

Phase 3 Evidence — Catalog Item Variables

21. ServiceNow Implementation Evidence

The following screenshots are appended to the Phase 3 document as implementation evidence. They correspond directly to the catalog item, variable configuration, passenger-facing catalog form, and QR widget activities described in the Phase 3 material.

Figure 21. Catalog Item Variables — Book A Metro Ticket



ServiceNow Maintain Items view showing the eight configured variables: Starting From, Going To, Type of Journey, No Of Passengers, Amount for Single Journey, Amount Including Return, Mode of Payment and Enter Payment Mode.

Phase 3 Evidence — Book A Metro Ticket Configuration

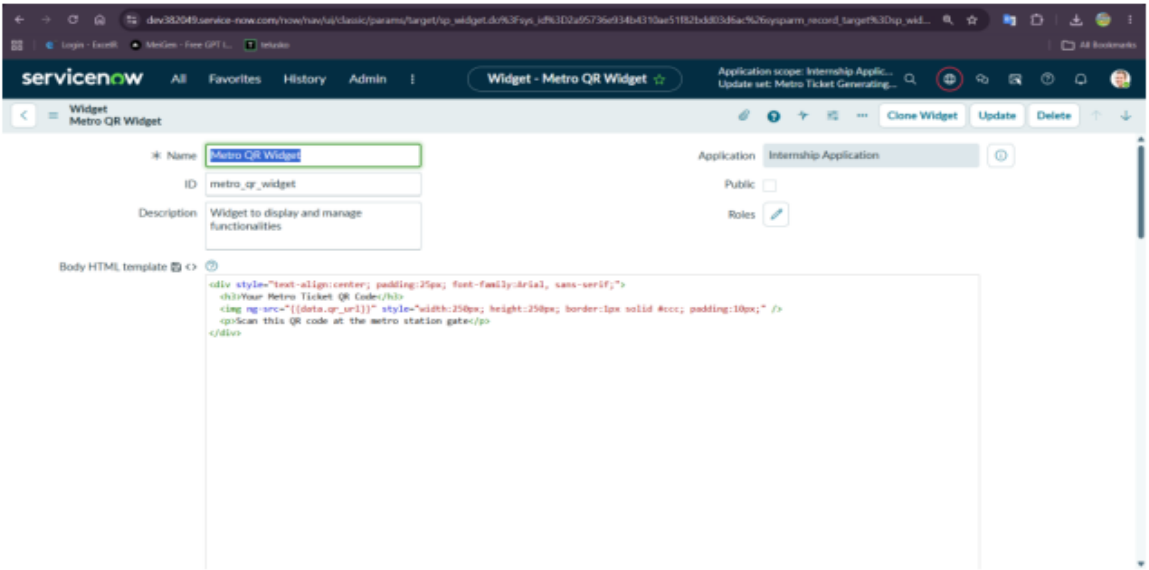
Figure 22. Catalog Item Configuration — Book A Metro Ticket

The screenshot shows the ServiceNow interface for configuring a catalog item. The browser address bar shows a URL from dev382049.service-now.com. The page title is 'Catalog Item - Book A Metro Ticket'. The breadcrumb trail is 'Catalog Item > Book A Metro Ticket'. The page has tabs for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A blue banner at the top says 'Build and modify items faster with the improved Catalog Builder.' Below this is a blue box with instructions: 'Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies. • Enter a Name and Short description to display for the item. • Enter a Price, approvals, variables, and other information as needed.' The form fields are: Name (Book A Metro Ticket), Application (Internship Application), Catalogs (Service Catalog), Category (Services), State (None), Checked out (None), Owner (System Administrator), and Secondary owners. The 'Active' checkbox is checked. The 'Fulfillment automation level' is set to 'Unspecified'. At the bottom, there are tabs for 'Item Details', 'Process Engine', 'Picture', 'Pricing', and 'Portal Settings'. The 'Short description' field contains 'metro ticket'.

Catalog Item form showing the item name, Service Catalog, Services category, active state, application scope and the metro ticket short description.

Phase 3 Evidence — Service Portal Entry Point

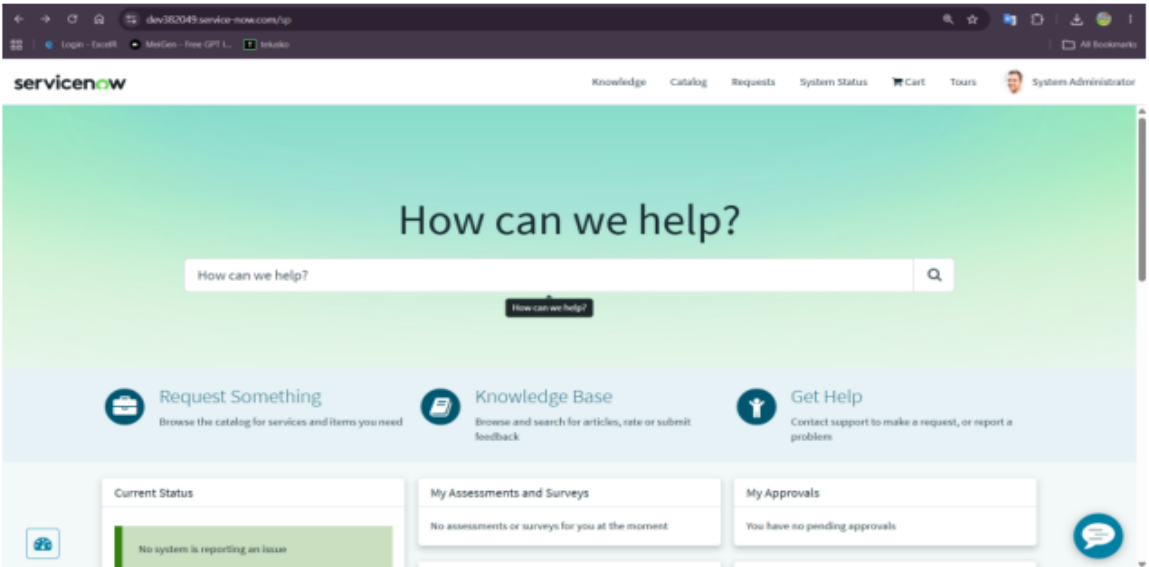
Figure 23. Service Portal Home — Passenger Entry Point



ServiceNow Service Portal landing page providing the passenger-facing interface and navigation to catalog services.

Phase 3 Evidence — Passenger Booking Form

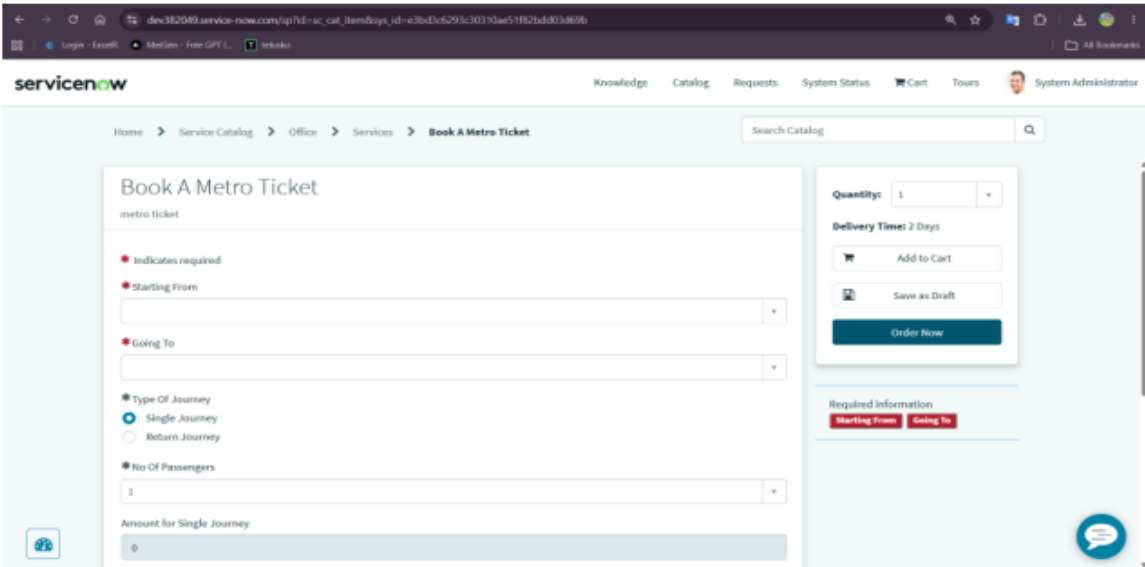
Figure 24. Book A Metro Ticket — Journey Details



Passenger-facing catalog form showing required Starting From and Going To fields, Single/Return journey selection, passenger count and calculated fare fields.

Phase 3 Evidence — Fare and Payment Section

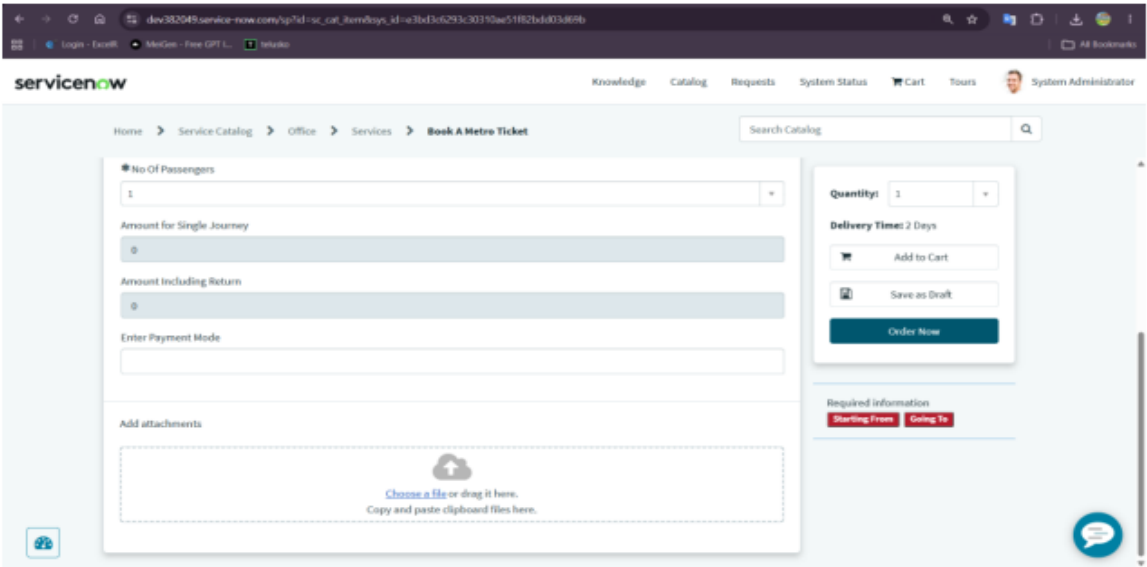
Figure 25. Book A Metro Ticket — Fare and Payment Section



Lower portion of the catalog form showing passenger count, fare outputs, Enter Payment Mode and checkout controls.

Phase 3 Evidence — Metro QR Widget

Figure 26. Metro QR Widget Configuration



ServiceNow widget record for Metro QR Widget, including the widget ID and Body HTML template that displays the generated QR image and scan instruction.

Phase 4 Evidence — Calculated Fare and QR Ticket

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22. Test Evidence — Calculated Fare

Figure 2. Catalog form showing dynamically populated fare fields.

The screenshot demonstrates the fare fields populated in the catalog interface. During QA, the value should be checked against the configured route, journey type and number of passengers.

23. Test Evidence — QR Ticket

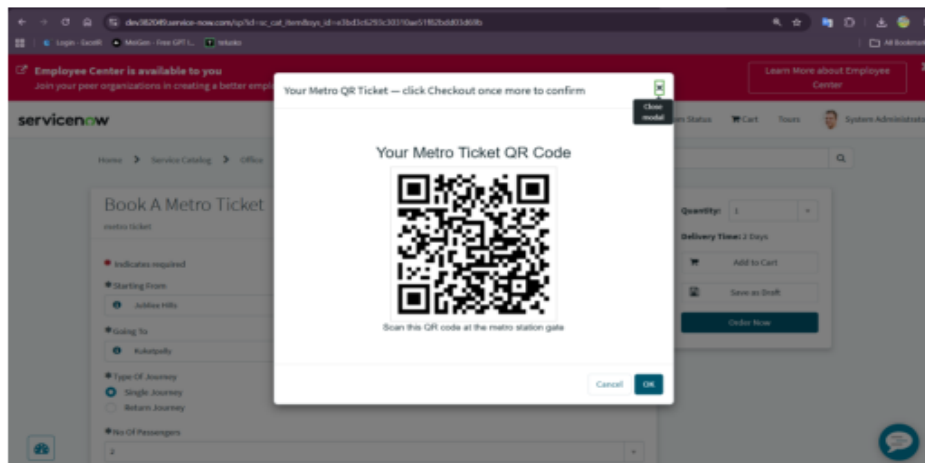


Figure 3. QR ticket modal displayed during the checkout flow.

The QR modal provides the final user-facing ticket experience. QA should confirm that it opens at the intended submission point, displays a valid QR image and does not bypass required validation or authorization.

24. Phase 4 Exit Criteria

- Access controls are reviewed and tested for intended roles.
- Catalog users can complete approved business actions without unnecessary database access.
- Mandatory and validation rules prevent incomplete or invalid submissions.
- Fare calculation is consistent across supported routes and journey types.
- UI policies and client scripts behave correctly after value changes.
- Field mapping and Process Automation populate the intended backend values.
- QR-ticket generation works through the intended checkout flow.
- Positive, negative and boundary QA scenarios are documented and passed.
- Data integrity checks confirm consistent master and transactional data.
- The solution is ready for Phase 5: Deployment, Documentation & Final Presentation.

25. Final Outcome

Phase 4 establishes the Metro Ticket Generation System as a validated, controlled and secure ServiceNow solution. The combination of ACL review, QA testing, validation, auto-population and field mapping ensures that the user experience is reliable while backend data

Phase 5 Evidence — Supplied Project Screenshots

17. Visual Evidence — Supplied Project Screenshots

The supplied screenshots are included as implementation evidence for the final documentation and presentation.

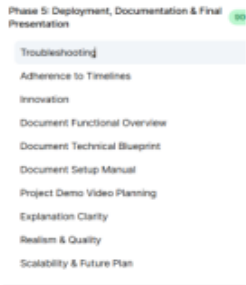


Figure 1. Supplied implementation evidence.

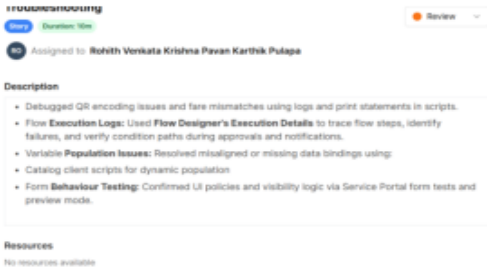


Figure 2. Supplied implementation evidence.



Figure 3. Supplied implementation evidence.

18. Explanation Clarity

Documentation and presentation follow a logical What → Why → How → Outcome structure. Each major step should have a clear heading, labelled evidence and a short explanation of why the configuration matters.

- Explain business purpose before technical configuration.
- Explain dependencies before demonstrating dependent features.
- Use consistent names for variables, tables and automation components.
- Show actual outcomes after configuration, not only setup screens.

Phase 5 Evidence — Additional Visual Evidence

19. Additional Visual Evidence

The next set of supplied screenshots provides evidence for configuration, workflow and project presentation.



Figure 4. Supplied project evidence.



Figure 5. Supplied project evidence.



Figure 6. Supplied project evidence.



Figure 7. Supplied project evidence.

20. Realism & Quality

The solution reflects a realistic city-metro workflow with station references, passenger-based fare calculation, journey types, payment selection, backend processing and QR-ticket presentation. Realistic sample routes and data make the demonstration meaningful.

- Use realistic station names and route combinations.
- Validate amounts against the configured fare matrix.
- Test normal and exception paths.
- Ensure the demo reflects the documented implementation.

Phase 5 Evidence — Presentation and Quality Evidence

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21. Final Evidence & Presentation Checklist

The remaining supplied screenshots are retained as final supporting evidence. They can be referenced during the final presentation to prove implementation and user-flow behavior.



Figure 8. Supplied project evidence.



Figure 9. Supplied project evidence.



Figure 10. Supplied project evidence.



Figure 11. Supplied project evidence.

22. Presentation Quality Checklist

- Start with the business objective and user problem.
- Explain the architecture before deep technical details.
- Demonstrate one complete successful booking.
- Show dynamic fare calculation and conditional payment behavior.
- Show the QR-ticket experience.
- Explain troubleshooting and security decisions.
- Finish with measurable outcome and future roadmap.

23. Scalability & Future Plan

- Extend SMS ticketing for confirmations and travel alerts.
- Add Email Ticketing for digital receipts and ticket delivery.
- Integrate a payment gateway.
- Introduce a travel analytics dashboard for route usage, passenger volume and revenue.
- Move route/fare logic toward maintainable data-driven fare tables and server-side calculation.